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GTA

Numbers don't always tell the whole story

By **Jennifer Yang** Staff Reporter

🚨 Sat., Jan. 30, 2010 | ⌚ 3 min. read

Scary numbers have dominated Toronto headlines this month: Seven dead in seven days. Fourteen pedestrians killed across the GTA. The deadliest January for city pedestrians in a decade.

There is a tendency to grip these disturbing numbers and wring them for meaning: Why such a spate of senseless tragedy in such a short time? What's causing the spike? Is January the deadliest month for crossing the road?

But for statisticians, maybe there *is* no meaning behind the numbers, just probability.

"Chances of a big clump are more than you would think," said Jeffrey Rosenthal, a professor of statistics at the University of Toronto. "Yes, (the numbers) are rare and January was certainly much worse than usual, but it wasn't something that is completely unexpected."

Plane crashes, forest fires, lightning strikes – for probability theorists, it's no surprise that such rare occurrences often happen in waves. Rosenthal, who authored *Struck by Lightning: The Curious World of Probabilities*, explains that random events are subject to a statistical phenomenon called Poisson bursts, or Poisson clumping, named for the 19th-century mathematician Siméon Denis Poisson.

The theory suggests that random events have a statistical tendency to occur in groups and Poisson bursts have been used to explain many a strange phenomenon. In Florida in 2001, for example, six surfers were attacked by sharks during one August weekend. Stranger yet, between November 2001 and March 2002, 11 prominent microbiologists died mysteriously.

Rosenthal said Poisson bursts probably were at work in November 2003, when there were five homicides in Toronto in one week.

"You might think that (random events) would be sort of equally spread out, so each month would have the same number of pedestrian deaths," Rosenthal said. "But that's not actually the way it works."

Seven isn't that big a number when examined through a statistician's lens. Rosenthal calculates between 2000 and 2009, Toronto averaged 31.9 pedestrian deaths a year and 2.7 deaths a month.

Using Poisson distribution, this means there is about a 1.9 per cent chance that seven or more pedestrian would die in a single month.

"You might expect that to happen every month in 50. So, once in four years, you might expect that there happens to be a month with seven deaths," Rosenthal said. "That makes it sound like it's not even that surprising," he added.

Toronto has seen spikes in roadside tragedies before. In January 2006, there were five deaths in a month, and in October 2005, eight pedestrians were killed. "Toronto has witnessed unexplained, short-term spikes in both death and injury collisions in the past many times," Toronto police Supt. Earl Witty said at a news conference Wednesday. "Our belief is this is a spike once again, only with more attention, as our neighbours in the GTA have seen the same collisions happening in their jurisdictions."

For Mike Brady, manager of traffic safety for the city's transportation services, seven deaths is tragic in any context, but statistically the number isn't very relevant.

"We find that about every 10 years, there seems to be an abnormally high number of pedestrian fatalities in January and February. I don't think it's the beginning of a new trend. I think it's an anomaly."

It's possible the rest of 2010 could be pedestrian fatality-free, said Const. Scott Parrish, of traffic services. "You can never pay too much attention to the numbers, but you can never ignore them. You just have to put them in context."

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